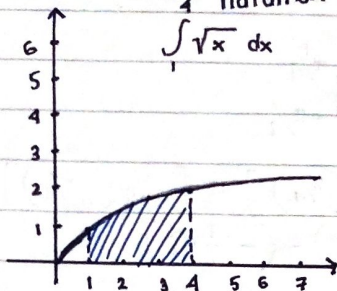
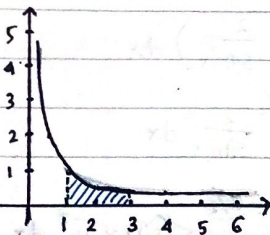


Soal Latihan 8.5 halaman 271

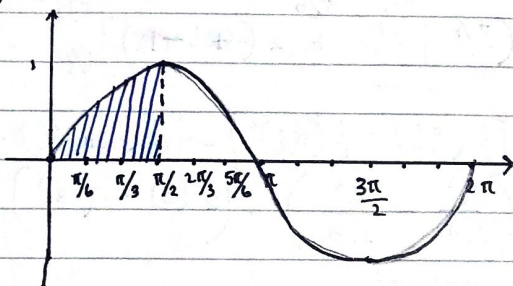
12. a.



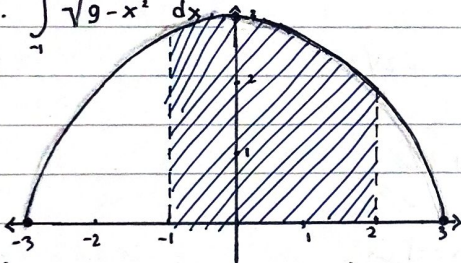
b. $\int_1^3 \frac{1}{x} dx$



c. $\int_0^{\pi/2} \sin x dx$



d. $\int_{-1}^2 \sqrt{9-x^2} dx$



16. $\int_0^2 (2-4x) dx = [2x - 2x^2]_0^2$
 $= [2(2) - 2(2)^2] - [2(0) - 2(0)^2]$
 $= [4 - 8] - [0]$
 $= -4$

17. $\int_0^3 |x-2| dx = \begin{cases} x-2, & x \geq 2 \\ 2-x, & x < 2 \end{cases}$

$\rightarrow \int_0^2 (2-x) dx + \int_2^3 (x-2) dx$
 $= [2x - \frac{1}{2}x^2]_0^2 + [\frac{1}{2}x^2 - 2x]_2^3$
 $= [(2(2) - \frac{1}{2}(2)^2) - (2(0) - \frac{1}{2}(0)^2)] + [(\frac{1}{2}(3)^2 - 2(3)) - (\frac{1}{2}(2)^2 - 2(2))]$

$= [(4-2) - 0] + [\frac{9}{2} - 6 - (2-4)]$

$= 2 + [-\frac{3}{2} + 2]$

$= \frac{5}{2}$

19). $\int_0^2 \sqrt{4-x^2} dx$

$\rightarrow$ merupakan Luas daerah di bawah $y = \sqrt{4-x^2}$ pada selang $[0, 2]$ $\rightarrow$ Pada kuadran I yang dibatasi sumbu koordinat & lingkaran $x^2 + y^2 = 4$
 Luas $\rightarrow \frac{1}{4}$ Luas seluruh lingkaran

$\rightarrow \frac{1}{4} \pi \cdot r^2 = \frac{1}{4} \pi (2)^2$
 $= \pi$

20). $\int_{-1}^1 \sqrt{1-x^2} dx$

$L = \frac{1}{2}$ Luas lingkaran $x^2 + y^2 = 1$
 $= \frac{1}{2} \pi \cdot (1)^2$
 $= \frac{1}{2} \pi$

21). $\int_{-3}^0 (2 + \sqrt{9-x^2}) dx$
 $= \int_{-3}^0 2 dx + \int_{-3}^0 \sqrt{9-x^2} dx$
 $= 2x \Big|_{-3}^0 + \frac{1}{4} \cdot \pi (3)^2$

$= 6 + \frac{9}{4} \pi$

22). $\int_0^{10} (2 + \sqrt{10x-x^2}) dx$
 $= \int_0^{10} 2 dx + \int_0^{10} \sqrt{10x-x^2} dx$

* $\int_0^{10} \sqrt{10x-x^2} dx$ adalah luas daerah di bawah $y = \sqrt{10x-x^2}$ pada selang $[0, 10]$ dibatasi sumbu koordinat dan lingkaran

$\rightarrow$ pers lingkaran :

$y = \sqrt{10x-x^2-25+25}$
 $= \sqrt{-1(x^2-10x+25)+25}$
 $y = \sqrt{-1(x-5)^2+25}$

$y^2 = -(x-5)^2 + 25$

$\rightarrow (x-5)^2 + y^2 = 25 \rightarrow$ lingkaran pusat $(5, 0)$ dan $r = 5$

$$\int_0^{10} 2 \, dx + \int_0^{10} \sqrt{10x - x^2} \, dx$$

$$= 2x \Big|_0^{10} + \frac{1}{2} \cdot \pi \cdot 5^2$$

$$= [20 - 0] + \frac{25}{2} \pi$$

$$= 20 + \frac{25}{2} \pi$$

$$25) \int_0^3 f(x) \, dx \rightarrow f(x) = \begin{cases} 2x, & x \leq 1 \\ 2, & x > 1 \end{cases}$$

$$= \int_0^1 f(x) \, dx + \int_1^3 f(x) \, dx$$

$$= \int_0^1 2x \, dx + \int_1^3 2 \, dx$$

$$= x^2 \Big|_0^1 + 2x \Big|_1^3$$

$$= [1^2 - 0^2] + [2(3) - 2(1)]$$

$$= 1 + (6 - 2)$$

$$= 1 + 4$$

$$= 5 //$$

► Soal Latihan 8.6

$$16) \int_1^4 \left(\frac{3}{\sqrt{x}} - 5\sqrt{x} - x^{-3/2} \right) dx$$

$$\int_1^4 \left(3x^{-1/2} - 5x^{1/2} - x^{-3/2} \right) dx$$

$$= \left[6x^{1/2} - \frac{10}{3}x^{3/2} - 2x^{-1/2} \right]_1^4$$

$$= \left[6\sqrt{x} - \frac{10}{3}x\sqrt{x} - \frac{2}{\sqrt{x}} \right]_1^4$$

$$= \left[6\sqrt{4} - \frac{10}{3}4\sqrt{4} - \frac{2}{\sqrt{4}} \right] - \left[6\sqrt{1} - \frac{10}{3}1\sqrt{1} - \frac{2}{\sqrt{1}} \right]$$

$$= \left[12 - \frac{80}{3} - 1 \right] - \left[6 - \frac{10}{3} - 2 \right]$$

$$= \left[11 - \frac{80}{3} \right] - \left[4 - \frac{10}{3} \right]$$

$$= 7 - \frac{70}{3}$$

//

$$19) \int_{-\pi/4}^{\pi/4} \cos x \, dx$$

$$= \left[-\sin x \right]_{-\pi/4}^{\pi/4}$$

$$= \left[-\sin\left(\frac{\pi}{4}\right) \right] - \left[-\sin\left(-\frac{\pi}{4}\right) \right]$$

$$= \left(-\frac{1}{2}\sqrt{2} \right) - \left(-\frac{1}{2}\sqrt{2} \right)$$

$$= 0 //$$

$$21) \int_{\pi/6}^{\pi/2} \left(x + \frac{2}{\sin^2 x} \right) dx$$

$$\int_{\pi/6}^{\pi/2} x \, dx + \int_{\pi/6}^{\pi/2} \frac{2}{\sin^2 x} \, dx$$

$$\int_{\pi/6}^{\pi/2} x \, dx + \int_{\pi/6}^{\pi/2} 2 \operatorname{cosec}^2 x \, dx$$

$$= \left[\frac{1}{2}x^2 \right]_{\pi/6}^{\pi/2} + 2 \left[-\cotan x \right]_{\pi/6}^{\pi/2}$$

$$= \left[\frac{1}{2} \left(\frac{\pi}{2} \right)^2 - \frac{1}{2} \left(\frac{\pi}{6} \right)^2 \right] - \left[\left(2 \cotan \frac{\pi}{2} \right) - \left(2 \cotan \frac{\pi}{6} \right) \right]$$

$$= \left(\frac{\pi^2}{2 \cdot 4} - \frac{\pi^2}{2 \cdot 36} \right) - \left(2(0) - 2(\sqrt{3}) \right)$$

$$= \frac{9\pi^2 - \pi^2}{72} + 2\sqrt{3}$$

$$= \frac{8\pi^2}{72} + 2\sqrt{3}$$

$$= \frac{\pi^2}{9} + 2\sqrt{3}$$

$$//$$

$$23) \int_0^2 |2x-3| \, dx$$

$$|2x-3| = \begin{cases} 2x-3, & x \geq \frac{3}{2} \\ 3-2x, & x < \frac{3}{2} \end{cases}$$

$$\int_0^2 |2x-3| \, dx = \int_0^{3/2} (3-2x) \, dx + \int_{3/2}^2 (2x-3) \, dx$$

$$= \left[3x - x^2 \right]_0^{3/2} + \left[x^2 - 3x \right]_{3/2}^2$$

$$\rightarrow \left[3 \left(\frac{3}{2} \right) - \left(\frac{3}{2} \right)^2 \right] - \left[\left((2)^2 - 3(2) \right) - \left(\left(\frac{3}{2} \right)^2 - 3 \left(\frac{3}{2} \right) \right) \right]$$

$$= \left[\frac{9}{2} - \frac{9}{4} \right] - \left[(4-6) - \left(\frac{9}{4} - \frac{9}{2} \right) \right]$$

$$= \frac{9}{4} - \left[-2 - \frac{9}{4} \right]$$

$$= 2 + \frac{9}{2}$$

//

Soal Latihan 8.7

1. f) $\int_0^3 (x+2)(x-3)^{20} dx$; $u = x-3$

misal : $u = x-3$

$du = dx$

• Batas bawah $\rightarrow u = x-3$

$u = -3$

• Batas atas $\rightarrow u = x-3$

$u = 0$

$\int_{-3}^0 (x+2) \cdot u^{20} du$

//

2. $\int_1^2 (4x-2)^3 dx$

• Cara 1

$u = 4x-2$

$= \int_1^2 (4x-2)^3 dx$

$du = 4 dx \rightarrow dx = \frac{du}{4}$

$= \int_1^2 (4x-2)^3 \frac{du}{4}$

$= \frac{1}{4} \cdot \frac{1}{4} (4x-2)^4 \Big|_1^2$

$= \left[\frac{1}{16} \cdot 6^4 \right] - \left[\frac{1}{16} \cdot 2^4 \right]$

$= \frac{1296}{16} - 1$

$= 81 - 1$

$= 80$

• Cara 2 //

$u = 4x-2$

$= \int_1^2 (4x-2)^3 dx$

$du = 4 dx \rightarrow dx = \frac{du}{4}$

• Batas bawah : $4(1)-2$

$= 2$

• Batas atas : $4(2)-2$

$= 6$

$\rightarrow \int_2^6 u^3 \frac{du}{4}$

$= \frac{1}{4} \cdot \frac{1}{4} u^4 \Big|_2^6$

$= \frac{1}{16} [6^4 - 2^4]$

$= \frac{1}{16} [1296 - 16]$

$= \frac{1}{16} [1280]$

$= 80$

5) $\int_{-5}^0 x \sqrt{4-x} dx$

Cara I: misal : $u = 4-x$

$x = 4-u$

$du = -dx$

$dx = -du$

$\int_{-5}^0 (4-u) \sqrt{u} du$

$= \int_{-5}^0 4\sqrt{u} - u\sqrt{u} du$

$= - \left[\int_{-5}^0 4\sqrt{u} - \int_{-5}^0 u\sqrt{u} du \right]$

$= - \left[\left(\frac{8}{3} u\sqrt{u} \right) \Big|_{-5}^0 - \frac{2}{5} u^2 \sqrt{u} \Big|_{-5}^0 \right]$

$= - \left[\left(\frac{8}{3} (4-x) \sqrt{4-x} \right) \Big|_{-5}^0 - \frac{2}{5} (4-x)^2 \sqrt{4-x} \Big|_{-5}^0 \right]$

$= - \left[\left(\frac{8}{3} \cdot 4 \sqrt{4} - \frac{8}{3} \cdot 9 \sqrt{9} \right) - \left(\frac{2}{5} (4)^2 \sqrt{4} - \frac{2}{5} (9)^2 \sqrt{9} \right) \right]$

$= - \left[\left(\frac{64}{3} - 72 \right) - \left(\frac{64}{5} - \frac{486}{5} \right) \right]$

$= - \left[\left(\frac{64}{3} - 72 \right) - \left(-\frac{422}{5} \right) \right]$

$= - \left[\left(-\frac{152}{3} \right) + \left(\frac{422}{5} \right) \right]$

$= \frac{152}{3} - \frac{422}{5}$

$= \frac{760}{15} - \frac{1266}{15}$

$= -\frac{506}{15}$

$= -\frac{506}{15}$

$= -33 \frac{11}{15}$

//

• Cara II : misal : $u = 4-x$

$x = 4-u$

$du = -dx$

Batas Bawah : $u = 4 - (-5)$

$= 9$

Batas Atas : $u = 4 - 0$

$= 4$

$\rightarrow \int_9^4 (4-u) \sqrt{u} du$

$= - \left[\int_9^4 4\sqrt{u} - \int_9^4 u\sqrt{u} du \right]$

$= \int_9^4 u\sqrt{u} du - \int_9^4 4\sqrt{u} du$

$$\begin{aligned}
 & \left[\frac{2}{5} u^{5/2} \right]_9 - \left[\frac{8}{3} u \sqrt{u} \right]_9 \\
 &= \left[\frac{2}{5} u^2 \sqrt{u} \right]_9 - \left[\frac{8}{3} u \sqrt{u} \right]_9 \\
 &= \left[\frac{2}{5} \cdot 16 \cdot 2 - \frac{2}{5} \cdot 81 \cdot 3 \right] - \left[\frac{8}{3} \cdot 4 \cdot 2 - \frac{8}{3} \cdot 9 \cdot 3 \right] \\
 &= \left(\frac{64}{5} - \frac{486}{5} \right) - \left(\frac{64}{3} - 72 \right) \\
 &= -\frac{422}{5} - \left(-\frac{152}{3} \right) \\
 &= -\frac{1266 + 760}{15} \\
 &= -33 \frac{8}{15} //
 \end{aligned}$$

8. $\int_0^{\pi/2} 4 \sin\left(\frac{1}{2}x\right) dx$

° Cara I :

= misal : $u = \frac{x}{2} \rightarrow x = 2u$

$\frac{\pi/2}{0} \quad dx = 2 du$

$\rightarrow \int_0^{\pi/2} 4 \cdot 2 \cdot \sin u \, du$

$= 8 \int_0^{\pi/2} \sin u \, du$

$= 8 \left(-\cos u \right) \Big|_0^{\pi/2}$

$= 8 \left(-\cos\left(\frac{x}{2}\right) \right) \Big|_0^{\pi/2}$

$= 8 \left(-\cos \pi/4 + \cos 0^\circ \right)$

$= 8 \left(-\frac{1}{2}\sqrt{2} + 1 \right)$

$= -4\sqrt{2} + 8 //$

Cara II :

misal : $u = \frac{x}{2} \rightarrow x = 2u$

$dx = 2 du$

Batas bawah : $u = \frac{0}{2} = 0$

Batas atas : $u = \pi/4$

$\rightarrow 8 \int_0^{\pi/4} \sin u \, du$

$= 8 \left(-\cos u \right) \Big|_0^{\pi/4}$

$= 8 \left(-\cos \pi/4 + \cos 0^\circ \right)$

$= 8 \left(-\frac{1}{2}\sqrt{2} + 1 \right)$

$= -4\sqrt{2} + 8 //$

11). $\int_1^3 \sqrt{5x-1} \, dx$

misal : $u = 5x-1$

$du = 5 dx$

Batas bawah : $5-1$

$= 4$

Batas atas : $10-1$

$= 9$

$\rightarrow \int_4^9 \sqrt{u} \cdot \frac{du}{5}$

$= \frac{1}{5} \left[\frac{2}{3} u \sqrt{u} \right]_4^9$

$= \frac{1}{5} \cdot \frac{2}{3} \left(9\sqrt{9} - 4\sqrt{4} \right)$

$= \frac{2}{15} (27 - 8)$

$= \frac{2}{15} (19)$

$= \frac{38}{15}$

$= 2 \frac{8}{15} //$

13). $\int_0^1 \sqrt[3]{a+bx} \, dx \quad (b \neq 0)$

misal : $a+bx = u$

$b \, dx = du \rightarrow dx = \frac{1}{b} du$

$\rightarrow \frac{1}{b} \int_0^1 \sqrt[3]{u} \, du$

$= \frac{1}{b} \left(\frac{3}{4} u \sqrt[3]{u} \right) \Big|_0^1$

$= \frac{1}{b} \left(\frac{3}{4} (a+bx) \sqrt[3]{a+bx} \right) \Big|_0^1$

$= \frac{1}{b} \left(\frac{3}{4} (a+b) \sqrt[3]{a+b} - \frac{3}{4} a \sqrt[3]{a} \right)$

$= \frac{3(a+b) \sqrt[3]{a+b} - 3a \sqrt[3]{a}}{4b} //$

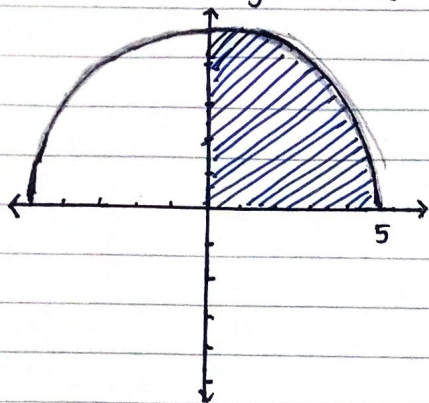
$$32. \int_0^{5/3} \sqrt{25 - 9x^2} dx, \quad u = 3x$$

$$du = 3dx \rightarrow dx = \frac{du}{3}$$

$$\rightarrow \int_0^5 \sqrt{25 - u^2} \frac{du}{3} \quad \left| \begin{array}{l} \text{BB} \rightarrow u = 0 \\ \text{BA} \rightarrow u = 5 \end{array} \right.$$

$$= \frac{1}{3} \int_0^5 \sqrt{25 - u^2} du$$

Berasal dari kurva $y = \sqrt{25 - u^2}$
 $y^2 + u^2 = 25$



$$\rightarrow \int_0^5 \sqrt{25 - u^2} du = L \frac{1}{4} \theta$$

$$= \frac{1}{4} \pi \cdot r^2$$

$$= \frac{1}{4} \pi \cdot 5^2$$

$$= \frac{25}{4} \pi //$$

$$\int_0^{5/3} \sqrt{25 - 9x^2} dx$$

$$\rightarrow \frac{1}{3} \cdot \frac{25}{4} \pi$$

$$= \frac{25}{12} \pi //$$

$$37. \int_{1/2}^1 \frac{1}{x^2} f(1/x) dx, \quad \text{jika} \int_1^2 f(x) dx = 3$$

$$\rightarrow \int_1^2 f(x) dx = 3$$

$$\left[\frac{1}{2} f^2(x) \right]_1^2 = 3$$

$$\cdot \frac{1}{2} f^2(2) - \frac{1}{2} f^2(1) = 3$$

misal : $u = 1/x$

batas bawah : 2

$$du = -x^{-2} dx$$

batas atas : 1

$$\rightarrow \int_2^1 \frac{1}{x^2} f(u) \cdot -du x^2$$

$$= \int_2^1 -f(u) du$$

$$= - \left[\frac{1}{2} f^2(u) \right]_2^1$$

$$= - \left(\frac{1}{2} f^2(1) - \frac{1}{2} f^2(2) \right)$$

$$= - (-3)$$

$$= 3 //$$